

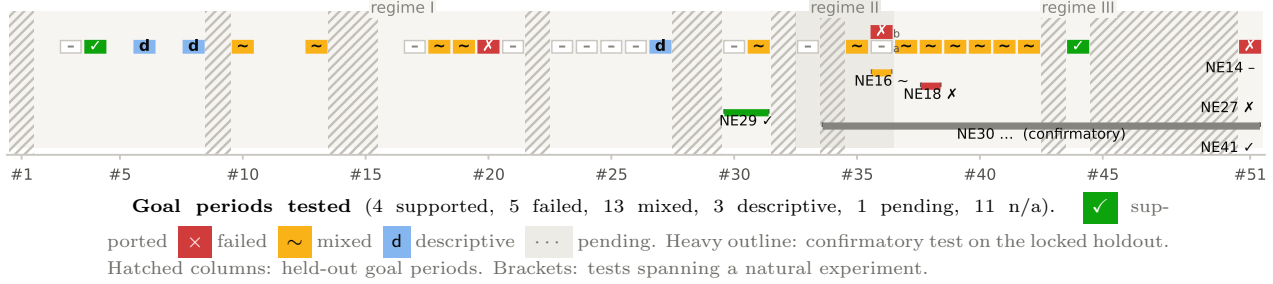
H15: Semantic information through natural scrambles: which information keeps agents and the swarm viable

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 55% Faithfulness 1.5/5 Usefulness 2.5/5

Bottom line. Day-scale memory (and the repo an agent works in) carries no detectable load-bearing information; the context window does: a forced wipe costs −39% of committed work for ten calls, about a tenth of output, whatever the agent saves to memory (round 1b; holdout pending).

level: hypothesis · pipeline checks 85% · updated 2026-10-04



WHAT HAPPENED

Natural scrambles in non-holdout data, per goal period, against a placebo-day null (a synthetic check on the real village skeleton: calibrated, power 17–41%). **Memory:** 12 losses of 50–80% of an agent’s memory and 20 newcomers with empty memories showed no day-scale cost on viability V^* : memory loss -0.55 SD $[-1.78, +0.68]$ (2 periods), newcomers $+0.15$ $[-0.18, +0.47]$ (*more engaged early*); the rewrite control was null. **Chat cuts:** only 3 usable events. **Context:** consolidations wipe the context window (forced every 41 turns) but keep memory. The pre-registered forced-vs-voluntary contrast came out wrong-signed (task-phase confound). Post hoc, writes drop 33–53% for ~10 turns after a forced wipe in 8/9 regime III periods, however much was saved to memory. D2.6 found no homeostatic V . **Round 1b** (work ledger, real failures, context ledger): the nulls stand and a forced wipe costs −39% work commits for ten calls (≈10% of output). Holdout not run.

PHYSICAL MODEL

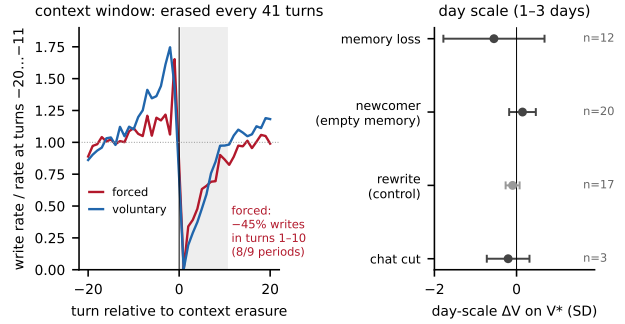
Model 04, Kolchinsky–Wolpert semantic information. X = one agent: memory M (day-scale store) and context window C (session-scale store, filled by chat and events); Y = the rest of the village. Scrambles erase M or C , or cut $Y \rightarrow X$; $\hat{p}$ = the scrambled joint. V = viability per agent-day (5 candidates) or per turn (writes); u = its same-day residual; δ = fraction removed; S , η = semantic information, its share of $I(X; Y)$.

$$\Delta V = V[p] - V[\hat{p}], \quad \kappa = \Delta V / \delta$$

$$S = \min_f I(\hat{p}^f) \text{ s.t. } V[\hat{p}^f] = V[p], \quad \eta = \frac{S}{I} \leq 1 - \delta^*$$

$$u_{ad} = V_{ad} - \bar{V}_{-a,d}$$

$$\widehat{\Delta V} = \bar{u}_{\text{post}} - (\mu_a + \rho^s \mathbb{E}[h_{t_0-1} | u_{\text{pre}}])$$



Observables. Left: write rate around consolidations, relative to turns $-20 \dots -11$ (regime III, pooled). A forced consolidation (41-turn cap, timing independent of task state) wipes the context: output falls ~45% for ten turns, then recovers (post-hoc). Voluntary ones dip alike after a write burst. Right: day-scale ΔV on V^* (meta over goal periods, 95% CI): memory loss, newcomers, rewrite control and chat cuts are all consistent with zero.

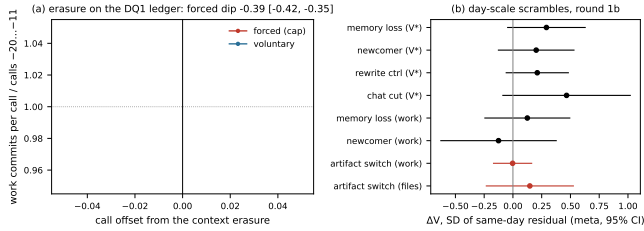
SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Mostly confirms what people running LLM agents already suspect: the working context keeps an agent productive, and a long-term memory file is something it rebuilds from its environment within a day. The one usable number is the cost of a context reset, about 45% of writes for ~10 turns (about a tenth of output at a 41-turn cadence). The levers are a longer cap or a task-state summary carried across the reset; today’s memory write doesn’t recover the loss. That is scaffold engineering, not physics, and post-hoc until the holdout check (C3) runs. Kolchinsky–Wolpert adds vocabulary (stored vs. observed information) but here measured neither S nor η , nor a viability function.

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RESULTS VS PREDICTIONS

Prediction	Round 1 → round 1b (im- proved data)	Verdict
D2.6 choose V^*	I: $V_{\text{rel}} \rightarrow V_{\text{eng}}$; III: $V_{\text{eng}} \rightarrow V_{\text{rel}}$ (real failures); V_{files} homeostatic (new)	inconcl.
P1 memory loss ≤ -0.25	$-0.55 [-1.78, 0.68] \rightarrow +0.29 [-0.05, 0.63]$	$\rightarrow \times$
P3 rewrite control null	$-0.10 \rightarrow +0.21$ ($z = -1.1 \rightarrow +1.5$)	$\checkmark$
P4 newcomers pay	$+0.15 \rightarrow +0.20$; progress $+0.31 [0.06, 0.56]$	$\times$
P5 forced < voluntary	writes $z + 4.9 \rightarrow \text{work } z + 2.3$	$\times$
P5' forced-wipe dip	writes -0.44 (8/9) $\rightarrow$ work $-0.39 [-0.42, -0.35]$ (7/9)	$\checkmark$ (NE41)
P6 chat cut harmless	$-0.72 \rightarrow -0.59$ (3 events)	$\times$
P11 artifact switch costs (new)	work $-0.00 [-0.17, 0.17]$; $\kappa_{\text{AS}} \approx 0 < \kappa_{\text{ML}}$	$\times$
NE41 memory dose (native)	$\rho = +0.01$; high-low tercile $-0.05 [-0.10, 0.01]$	$\checkmark$
NE29 lineage retired incumbents' (native)	work $z = +0.13$	$\checkmark$



Second figure. Round 1b. (a) Work commits per call around a context erasure on the DQ1 ledger, relative to calls $-20 \dots -11$: a forced wipe (41-turn cap) cuts committed work by about 39% for ten calls; voluntary consolidations dip more after a work burst (task phase). (b) Day-scale scrambles (meta over goal periods, 95% CI): memory loss, newcomers, the rewrite control, chat cuts and artifact switches, on V^* and on work output. Nothing but the chat cut (3 events) departs from zero.

GOAL PERIODS IN DETAIL

NE30	...	pending (confirmatory; not run) (holdout)
G04	$\checkmark$	1b: supported ($V^* V_{\text{eng}}$; MN -0.50 ($z = -0.7$); 2026-10-0...
G06	d	1b: descriptive ($V^* V_{\text{eng}}$; MR $+0.40$ ($z = +0.5$); 2026-10...
G08	d	1b: descriptive ($V^* V_{\text{eng}}$; MR -0.69 ($z = -1.0$); 2026-10...
G10	$\sim$	1b: mixed ($V^* V_{\text{eng}}$; MN $+0.49$ ($z = +1.5$); 2026-10-04, Γ ...
G13	$\sim$	1b: mixed ($V^* V_{\text{eng}}$; ML $+0.40$ ($z = +1.2$); 2026-10-04, Γ ...
G18	$\sim$	1b: mixed ($V^* V_{\text{eng}}$; MN $+0.58$ ($z = +1.2$); 2026-10-04, Γ ...
G19	$\sim$	1b: mixed ($V^* V_{\text{eng}}$; MN $+0.96$ ($z = +1.2$); 2026-10-04, Γ ...
G20	$\times$	1b: failed ($V^* V_{\text{eng}}$; MN $+0.56$ ($z = +1.1$); 2026-10-04, ...
G27	d	1b: descriptive ($V^* V_{\text{eng}}$; MR $+0.10$ ($z = +0.2$); 2026-10...

+27 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	0	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`witeup/paper.tex`).

CAVEATS

Day-scale tests stay underpowered (synthetic power 17–41%), and V^* moved in both regimes when the reliability measure was corrected, so period verdicts are replications under a changed yardstick. 118 of 172 artifact switches fall on a goal's first day and cannot be estimated within a period; the rest are agent-chosen, like voluntary consolidations. The erasure dip is post-hoc in origin; NE41 replicated it with dated predictions on independent data but the holdout (C3) has not run. V_{files} passed D2.6 on 5 shocks and is outside the pre-registered pool.

ROUND 2

Re-freeze `confirm_ne30.py` on work commits and run C3; pre-register V_{files} on new shocks.

- **Essence.** Where does the swarm keep its knowledge? In artifacts and history, not in agents.
- **H15-R1** Stigmergic memory: after a memory loss, agents re-acquire state from artifacts (file reads, history search) within a few turns; richer artifact trails mean faster recovery (E3).
- **H15-R2** Semantic information is concentrated in the few items read first after a reset (the goal and the last messages).
- **H15-R3** Memory notes are performative: their content rarely reappears in later actions.

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